Part 2: 41 Marks

Introduction

When using the ATAR Calculator to decide which University courses they wish to apply for, students need to predict their study scores in different subjects. In order to assist students to do this a team of Mathematicians at Killester College are analysing the relationship between a student's SAC results and their Final Study Score.

The SAC results for a random group of 18 students from Killester 2019 Further Mathematics classes are used to complete this investigation.

In this task we will investigate the following:

- **analyse a Year 12 Further Maths Class' students' SAC results.**
- **Query whether a relationship exists between the SAC results and the Final Study Scores?**
- **Conclude whether an accurate prediction of a student's Final Study Score in Further Mathematics can be made from their SAC results.**

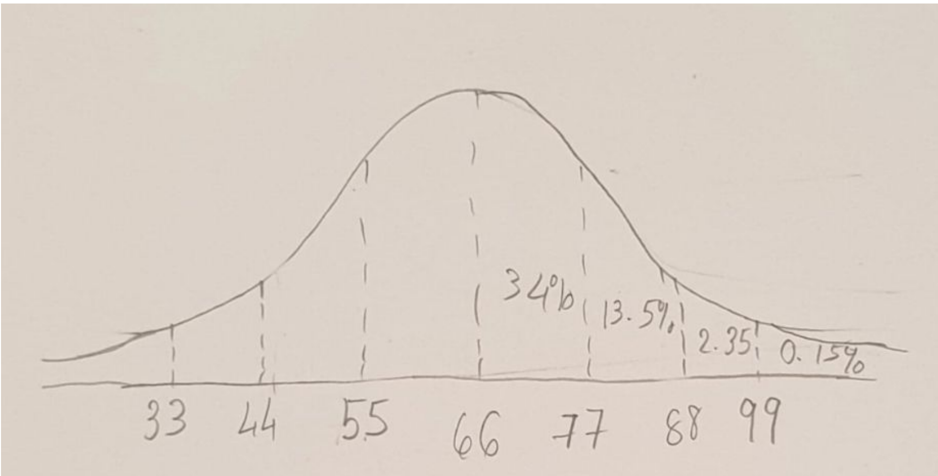
Note: For the purposes of this investigation, the SAC results refer to the mean of the students' Further Maths SACs for the year, and are out of 100.

The initial task of the investigation

Before investigating the relationship between the SAC results and the Final Study Scores, the team of Mathematicians at Killester decide that they want to revisit the previous analysis of a FM class' SAC results in 2018 to have a better understanding of how this group of Killester students performed in their SACs.

The team found that the SAC results for this class were normally distributed with a mean of 66 and a standard deviation of 11 (SAC results were out of 100 and there were 25 students in this class).

1. Draw the bell-shaped curve representing the normal distribution for the SAC results for this class. Include values three standard deviations from the mean. (2 marks)



SAC Results

2. There were 25 students in this class. How many students had achieved a result greater than 77? (1 mark)

$\frac{16}{100} \times 25 = 4$ students

3. Two students are selected with the Z-score of 2.11 and 2.43.

(a) Are these two students in the top 2.5% of the group. Give reasons why? (1 mark)

Yes, they are as both their z-scores are greater than 2, which means their SAC results are 2 standard deviations above the mean. So they are in the top 2.5% of the group.

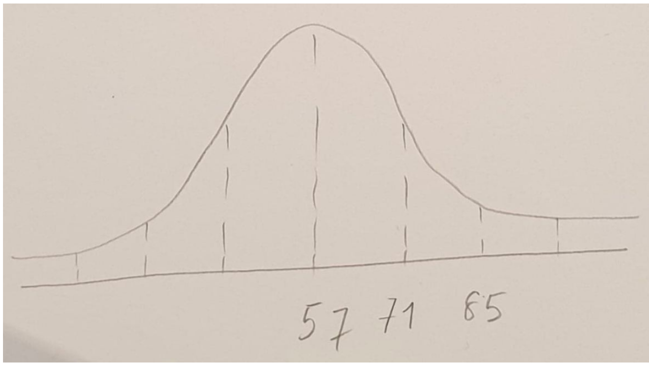
(b) Calculate their actual SAC results correct to 2 decimal places (2 marks)

student 1 : $\frac{x-66}{11} = 2.11 \Rightarrow x = 89.21$

Student 2: $\frac{x-66}{11} = 2.43 \Rightarrow x = 92.73$

(c) The SAC results for all students studying Further Mathematics in Victoria in 2018 are normally distributed with a mean percentage of 57 and a standard deviation of 14. In this 2018 group of Year 12 FM students at Killester, there were 5 students achieving a SAC result greater than 85.

Approximately what percentage of the class would be in the top 2.5% of the state ? (2 marks)



These 5 students are all in the top 2.5% of the state as their results are in 2 standard deviations above the mean of the state.

$\frac{5}{25} \times \mathbf{100 = 20\%}$

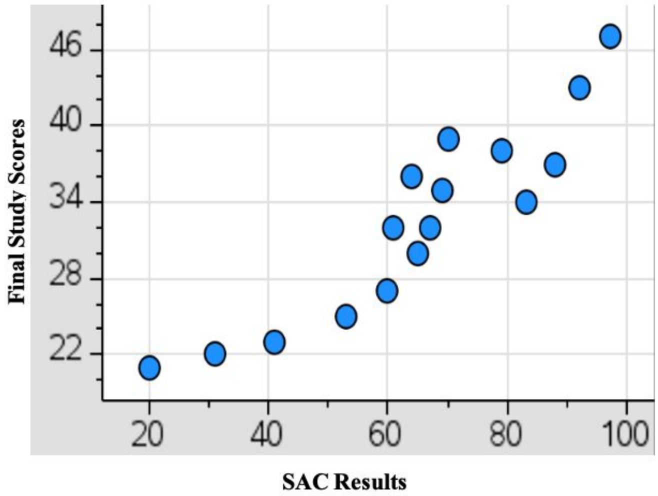
The next phase of the investigation

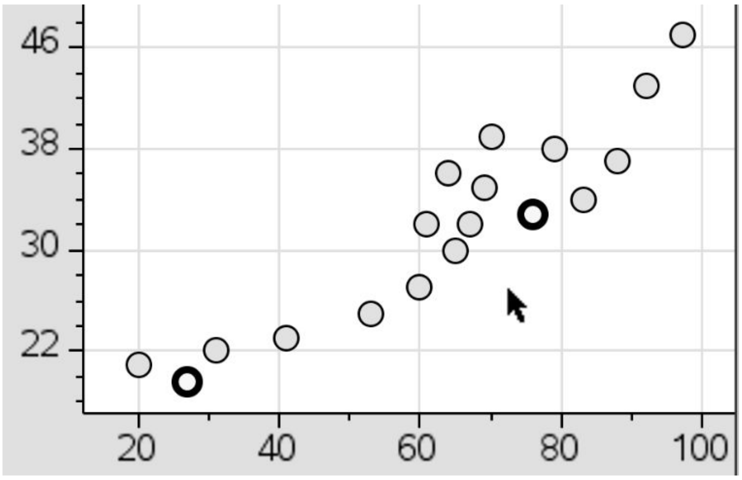
The team of Mathematicians at Killester has randomly selected a group of 18 students' SAC results and their final Study Score from 2019 Further Mathematics classes, shown below.

Student Results:

STUDENT	SAC RESULT Score out of 100	STUDY SCORE Score out of 50
A	20	21
B	27	20
C	31	22
D	53	25
E	41	23
F	61	32
G	67	32
H	60	27
I	70	39
J	64	36
K	65	30
L	69	35
M	79	38
N	76	33
O	83	34
P	88	37
Q	92	43
R	97	47

4. **A scatterplot of the data is provided but it is missing two pieces of data – student B and student N. Fill in their results on the scatterplot and label them clearly. (2 marks)**





5. **What is the form of this scatterplot? (1 mark)**

non-linear

6. **What would be the problem with fitting a linear regression model to this data? (1 mark)**

It wouldn't be a real representation of how the final score increases and therefore you would get incorrect/unreliable values if you were to predict the final scores using a linear model.

Correlation coefficient will not be accurate to measure the strength of the linear association.

7. **Which transformations should be considered in order to linearise this data? (1 mark)**

x^2 , $\frac{1}{y}$ and log(y)

8. Complete the following table using the transformations you chose in question 7. (3 marks)

Transformation	**Coefficient of determination** **(to 3 significant figures)**	**Equations in terms of SAC results and Study Score (to 3 significant figures)**
x^2	0.867	**Study Score = 19.6 + 0.00274 SAC Result^2**
$\frac{1}{y}$	0.900	$\frac{1}{Study\ Score}$ **= 0.0568 - 0.000371 SAC Result**
log(y)	0.889	**log (Study Score) = 1.20 + 0.00465 SAC Result**

9. Which transformation of data would provide the highest predicted Study Score for students? Explain your chosen model using student L's SAC result of 69. (2 marks)

Study Score = 19.6 + 0.00274 $\times$ 69^2

= 32.65

$\frac{1}{Study\ Score}$ **= 0.0568 + 0.000371 $\times$ 69**

Study score = 32.05

log (Study Score) = 1.20 + 0.00465 $\times$ 69

Study score = 33.18 . log(y) is the model that provides the highest predicted score.

The team of Mathematicians at Killester College has decided to choose the model below.

$$\textbf{Study score} = 19.6 + 0.00274 \times \textbf{SACs Result}^2$$

The model above will apply to the following **questions 10 - 14.**

10. Describe the association between the SAC results and the Study Scores in terms of strengths and direction. (1 mark)

Strong, positive

11. What percentage of the variation in the Study Score can NOT be explained by the variation in the SAC result. (give your answer correct to 1 decimal place) (1 mark)

100 - 0.867 *100 = 13.3%

12. A student who achieves 62 on SACs wants to use this model to predict her Study Score.

(a) Find her predicted study score, correct the nearest whole number. (1 mark)

$$19.6 + 0.00274 \times 62^2 = 30$$

(b) Comment on the reliability of this prediction. (1 mark)

It is reliable as the SAC result is within the range.

13. Allysa was aiming for a study score of 40. Using the provided model, what mean mark must she achieve on her SACs to the nearest whole number? (1 mark)

$40 = 19.6 + 0.00274 \times x^2$

$x = 86.29 \approx 86$

14. a) What is the residual for Student J to 2 decimal places? (1 mark)

Predicted final score = $19.6 + 0.00274 \times 64^2$

$= 30.82$

Residual = 36 - 30.82

= 5.18

b) What does this residual mean? (1 mark)

The data point is above the regression line.

That is, the actual final score of a student is approximately 5.18 more than we would predict

Establishing a Maths Enhancement Centre

To help support Further Mathematics students to do well in their SACs and the end of year examinations, a drop-in centre called Maths Enhancement Centre has been established. They provide one-on-one help with students out of school hours as well as provide the option to purchase summary topic books.

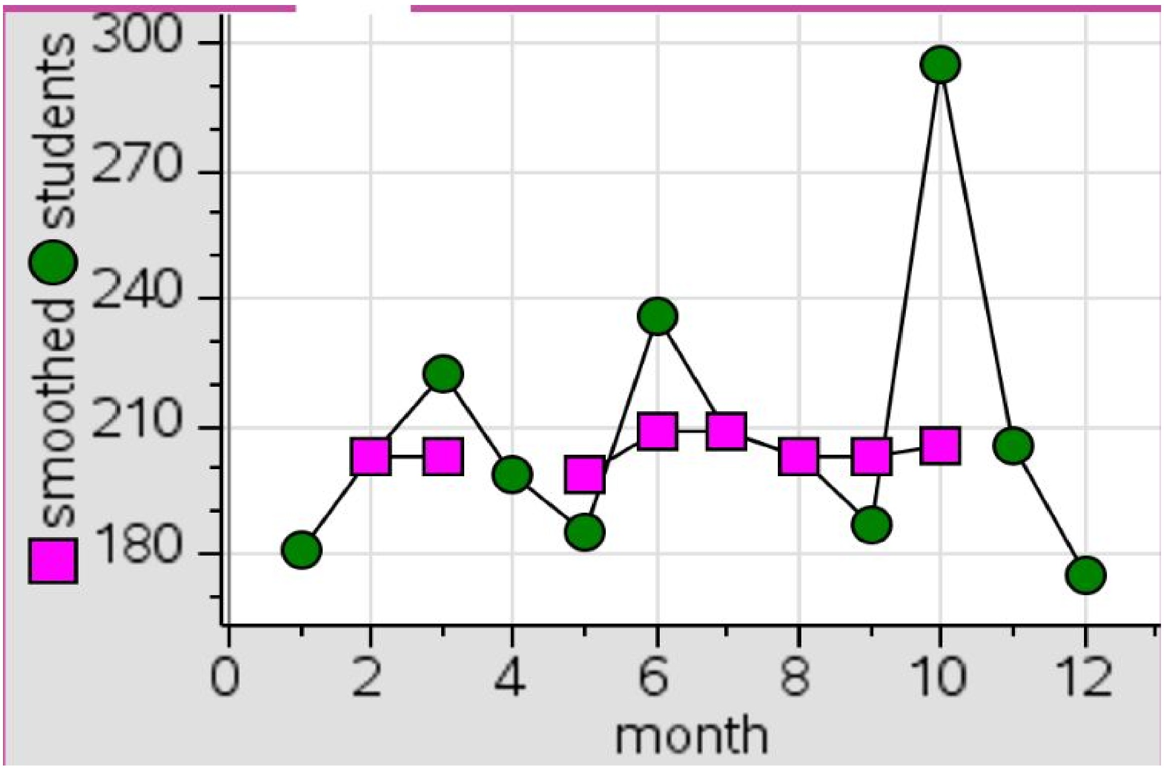
One of the supervisors believes there is an increasing trend in the number of students accessing the one-on-one help sessions and has been gathering data in 2019.

Month	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sep	Oct	Nov	Dec
Number of Students attending one-on-one sessions.	181	203	222	199	185	236	209	203	187	295	205	175

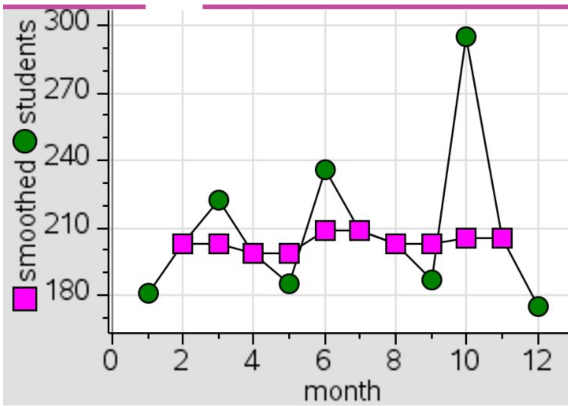
The time series plot for the 2019 data is provided below:

Raw data - the number of students time series

The three median smoothed number of students time series



15. Complete the three median smoothed time series plot on the above graph. (2 marks)



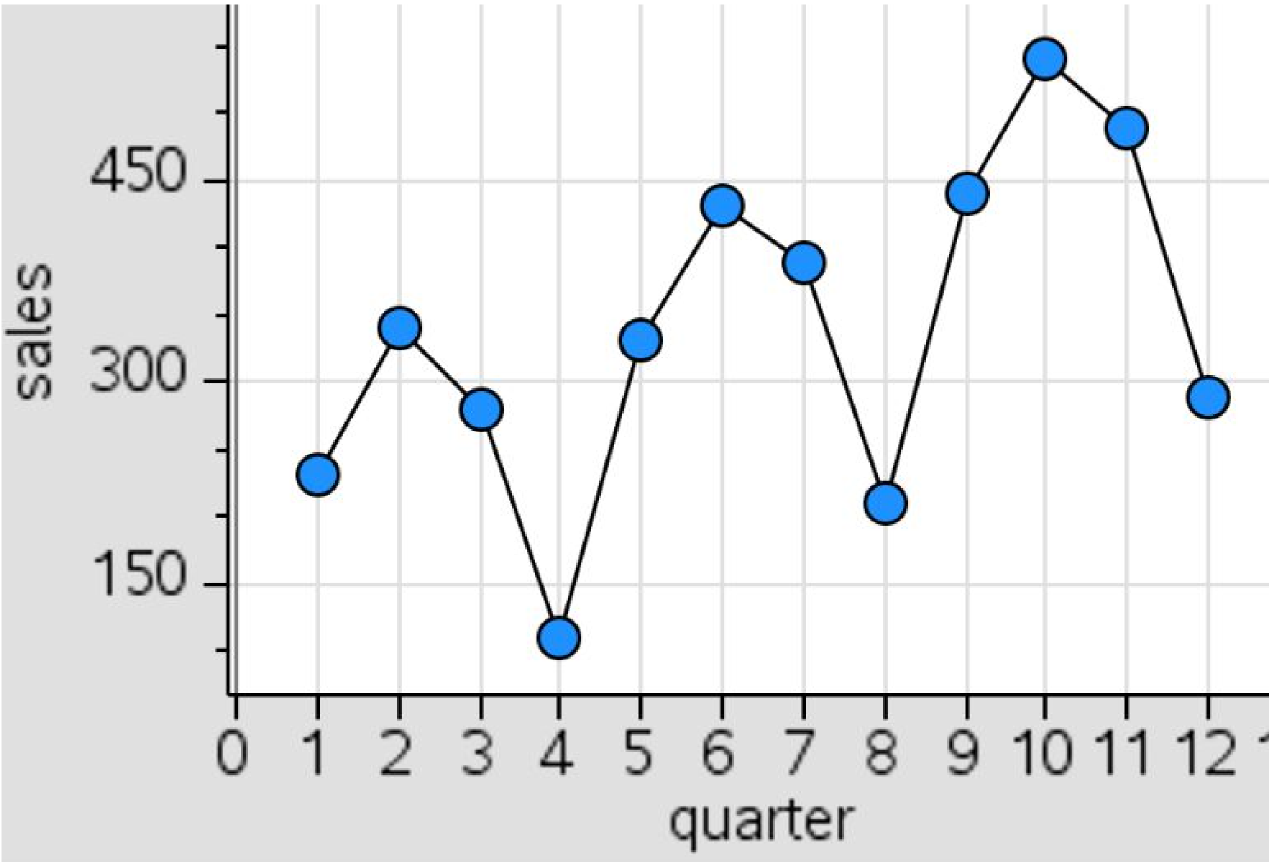
16. Does the three median smoothed time series support the supervisor's belief that there has been an increasing trend in the number of students accessing the one-on-one help sessions during the previous year? Briefly explain your answer. (1 mark)

No, the three median smoothed time does not support the supervisor's belief of an increasing trend. The smoothed time series data shows a flat/constant trend of the number of students accessing the one-on-one help.

The Maths Enhancement Centre has also been monitoring the sales of their topic summary books. The following table shows the number of sales of the topic summary books in each quarter over the last 3 years.

	Quarter 1	**Quarter 2**	**Quarter 3**	**Quarter 4**	**Quarterly Average**
2017	230	340	280	110	240
2018	330	430	390	210	**340**
2019	440	540	490	290	**440**
Seasonal Index	0.98		1.14	0.58	

The time series plot for this three year period is provided below.



17. Does this time series graph indicate any seasonality in the data? Explain. (1 mark)

Yes, there is seasonality in the data as there is periodic movements related to quarter. Each year quarter 2 has the highest sales and quarter 4 has the lowest sales.

18. Calculate the quarterly averages for 2018 and 2019. Put your answers in the table on the previous page. (1 mark)

2018: (230+340+280+110)/4 = 340

2019: (440+540+490+290)/4 = 440

19. Calculate the seasonal index for Quarter 2 correct to 2 decimal places. Interpret what this seasonal index means. (3 marks)

SI = 4 - (0.98+ 1.14 + 0.58)

SI = 1.30 (which means the sales for quarter 1 are 30% more than what is expected for an average quarter)

20. The following table shows the deseasonalised quarterly topic summary book sales. Find the missing value for Quarter 1, rounding to the nearest book. (1 mark)

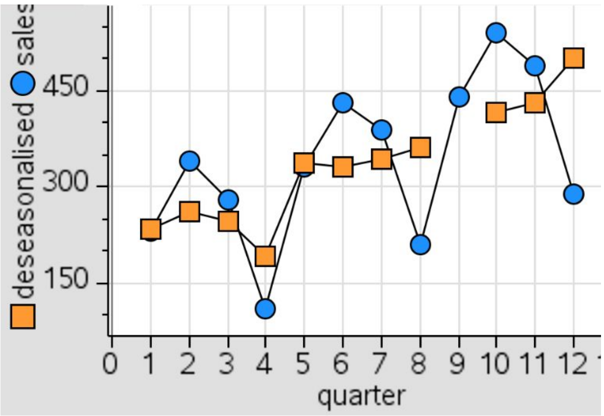
	Quarter 1	Quarter 2	Quarter 3	Quarter 4
2017	235	262	246	190
2018	337	331	342	362
2019		415	430	500

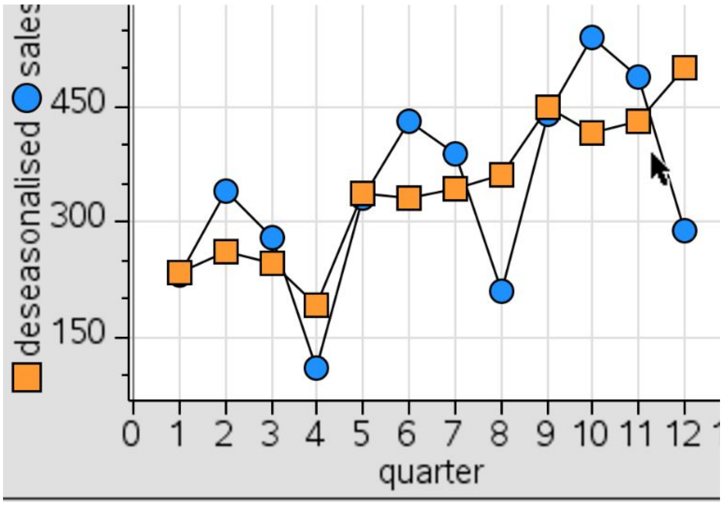
Deseaonalised sales $= \frac{440}{0.98}$

$= 448.98 \simeq 449$ **books**

21. Using your answer from question 20, complete the following time series of the 2017-2019 deseasonalised quarterly topic summary book sales. (1 mark)

Raw data- Sales of topic summary books **Deseasonalised Sales**





22. Comment on the trend displayed by the deseasonalised data. (1 mark)

The deseasonalised data has an upward trend

23. Determine the least squares regression equation for the deseasonalised quarterly topic summary book sales. Give your answers to two decimal places. (2 marks)

Deseasonalised topic summary book sales = ___**182.70**_ + ___**24.44**__ × quarter

24. Graph the deseasonalised equation from question 23 on your graph from question 21. Show the calculations for the two coordinate points used to construct the line and mark them in on the graph. (Give your answer to the nearest whole number) (2 marks)

$=182.70 + 24.44 \times 1$

$= 207.14$

$(1, 207)$

$= 182.70 + 24.44 \times 10$

$= 427.1$

$= (10, 427)$

25. Calculate the expected number of topic summary book sales for Quarter 3 in 2020. Show your working out. (Give your answer to the nearest whole number) (2 marks)

Quarter 3 in 2020 =15

Deseasonalised sales $= 182.70 + 24.44 \times 15$

$= 549.3$

Reseasonalised sales $= 549.3 \times 1.14$

$= 626.202 \simeq 626 \text{ books}$

END OF TEST - 41 Marks